

PROJECT DESIGN PHASE

Problem – Solution Fit Template

Metro Ticket Generating System

Date	Team ID	Project Name	Maximum Marks
04 September 2026	_____	Metro Ticket Generating System	2 Marks

Problem–Solution Fit Canvas

1. CUSTOMER SEGMENT(S) Metro passengers / customers who need to book tickets for a selected route and journey. Secondary stakeholders include station/metro operations personnel who need convenient ticket verification and IT administrators who configure and secure the ServiceNow application.	2. JOBS-TO-BE-DONE / PROBLEMS • Book a metro ticket by selecting the starting station, destination station, journey type and passenger count. • Know the applicable fare without manually calculating it. • Complete the request through a guided digital form. • Obtain convenient proof of booking that can be verified at the station. • Reduce errors and unnecessary steps in the ticket-booking process.
3. TRIGGERS • Need to travel on a metro route. • Need to book a ticket quickly before a journey. • Need to know the payable fare for the selected route and passenger count. • Need a digital ticket that can be presented and verified at the station.	4. EMOTIONS: BEFORE / AFTER Before: inconvenience, uncertainty about fare, concern about manual steps or unclear confirmation. After: confidence that the request contains the required information, clarity about the fare, and convenience from having a QR-based digital ticket.
5. AVAILABLE SOLUTIONS • Manual / traditional ticket booking at a station counter or ticketing facility. • Digital booking systems that require users to enter journey information and receive confirmation. • The proposed ServiceNow solution combines a Service Catalog request form, automated fare calculation, validation, QR generation and portal presentation.	6. CUSTOMER CONSTRAINTS • Limited time while preparing for a journey. • Users may not want to perform fare calculations manually. • Booking should not require direct access to backend database tables. • The interface should guide users toward required and valid information. • The solution should work within the available ServiceNow platform capabilities.
7. BEHAVIOUR • Selects source and destination stations. • Chooses single or return journey and enters passenger count. • Provides payment-related information where required. • Submits the Service Catalog request. • Uses the generated QR code as digital proof of the ticket.	8. CHANNELS OF BEHAVIOUR 8.1 ONLINE: ServiceNow Service Catalog / portal booking form, digital request submission, QR-code display and ticket information. 8.2 OFFLINE: Travel planning, reaching the station, presenting the digital QR ticket and station-side verification.

9. PROBLEM ROOT CAUSE

The booking experience can become inconvenient when journey information, fare calculation and ticket confirmation are handled as separate or manual activities. Users need a guided process that reduces repetitive entry, calculates the fare consistently and provides a verifiable digital ticket.

10. YOUR SOLUTION

A ServiceNow-based Metro Ticket Generating System that provides a guided Service Catalog booking experience. The solution uses route/station data, dynamic form behaviour, automatic fare calculation, validation, QR-code generation and a Service Portal widget. Role-based access controls protect backend data while allowing intended users to complete the booking journey.

References: Ideahackers Problem-Solution Fit Canvas; Medium Problem-Solution Fit Canvas.